# Project Planning & Management

Customer Churn Analysis and Retention Strategy for the Olist E-Commerce Marketplace

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|---|---|
| **Project title** | Customer Churn Analysis and Retention Strategy (Olist Brazilian E-Commerce dataset) |
| **Team name** | Customer Churn |
| **Track** | Data Engineering |
| **Team Leader** | Ziad Khaled Abd El-Rahman |
| **Team members (4 to 6)** | 1. Ziad Khaled Abd El-Rahman, 2. Omar Hassan Ibrahim, 3. Amr Mostafa Etman, 4. Malak Ashraf Mostafa, 5. Mennatullah Hesham Mohamed, 6. Ahmed Ayman Naguib |
| **Lecturer / Instructor** | Ahmed Essam Azab |
| **GitHub repository** | https://github.com/nhahub/NHA-5-086 |
| **Document date** | 30 October 2026 |
| **Document version** | 1.0 (submitted for lecturer review) |

## 1. Project Proposal

### 1.1 Overview

Customer churn means customers who stop buying from a business. This project analyzes customer behavior on Olist, a Brazilian marketplace that connects small sellers to customers, to find the patterns that lead to churn. We will build a simple, interactive dashboard that shows which customers churn and why, and use the findings to propose data-backed customer retention strategies.

The project is based on the Churn Analysis and Retention Strategy project in the program booklet. It focuses on data preparation, exploration and business insight, and presents the results in a simple dashboard instead of a trained prediction model.

### 1.2 Problem Statement

The Olist dataset has no churn column, and its data is split across 9 separate tables. The business therefore cannot see which customers are lost, why they leave, or which customers are at risk. Without this, it cannot decide where to spend money on retention (for example discounts, faster delivery or better service).

### 1.3 Dataset

**Source:** Olist Brazilian E-Commerce Public Dataset (Kaggle): https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

About 100,000 orders placed between 2016 and 2018, stored in 9 linked tables: orders, order items, payments, reviews, customers, sellers, products, geolocation and product category name translation.

### 1.4 Churn Definition (creating the target column)

The dataset has no churn label, so the team will create one. A customer is considered churned if they do not place another order within 90 days. The exact rules are below, so the label is clear, repeatable and easy to defend in the final presentation.

| Item | Rule |
|---|---|
| **Customer identifier** | customer_unique_id from the customers table. The customer_id column changes with every order, so using it would wrongly make every customer look like a one-time buyer. |
| **Orders counted** | Orders with status delivered. Canceled and unavailable orders are excluded. |
| **Order date** | order_purchase_timestamp |
| **Starting point (T0)** | The date of the customer's first delivered order. |
| **Target column: churn** | **churn = 1** if the customer places no further order within 90 days after T0. **churn = 0** if the customer places a second order within 90 days after T0. |
| **Observation window** | Customers whose first order is less than 90 days before the last order date in the dataset are removed from the analysis, because their 90 days cannot be fully observed. |
| **Dashboard breakdowns** | Churn is analyzed by information known from the first order: order value, freight cost, number of items, payment type and installments, product category, customer state, delivery delay and late-delivery flag, and review score. |
| **Sensitivity check (optional)** | Repeat the labeling with 60 and 180 days to show how the result changes with the chosen threshold. |

**Important expectation:** most Olist customers buy only once (commonly reported at around 97%), so most customers will have churn = 1. The team will confirm the exact class balance in the exploration step and show it clearly in the dashboard and report (see Risk 1).

### 1.5 Objectives

- Clean and combine the 9 tables into one reliable customer-level data model.
- Create the churn target column using the 90-day rule and document it.
- Explore the data to find what is linked to churn (for example late delivery, low review score, payment type, product category, state) and produce a class-imbalance report.
- Build a simple interactive dashboard (Power BI or Streamlit) with KPI cards, charts and filters that show the churn rate and its main drivers.
- Write a retention strategy report with actions backed by the dashboard findings.
- Deliver full documentation, testing, a user manual and the final presentation on time.

### 1.6 Scope

| In scope | Out of scope |
|---|---|
| Data cleaning, joining and churn labeling | Real-time or live data |
| Exploratory analysis, class-imbalance report and insights | Production deployment for real customers |
| Simple interactive churn dashboard (charts and filters) | Mobile app |
| Retention strategy report based on the dashboard | Paid tools or services |
| Documentation, testing, user manual, final presentation | Running real retention campaigns or A/B tests |
| Optional only: a simple churn prediction model | Using data outside the Olist dataset |

**Optional extra (only if time allows):** a simple churn prediction model (for example Logistic Regression) with its results shown in the dashboard, and the 60/180-day sensitivity analysis.

### 1.7 Deliverables

| Milestone | Deliverables | Planned date |
|---|---|---|
| **1. Data collection, exploration and preprocessing** | EDA notebook with visualizations (bar charts, pie charts, histograms); cleaned dataset with the churn column; summary report of key customer behavior insights | 23 Nov 2026 |
| **2. Dashboard and visualization** | Interactive churn dashboard (KPI cards, charts, filters); dashboard screenshots or live demo | 30 Nov 2026 |
| **3. Retention strategy and reporting** | Retention strategy report; demo presentation | 30 Nov 2026 |
| **Documentation and reports** | Requirements and design documents, GitHub repository with README, test cases and bug report, user manual, final presentation | 4 Dec 2026 |

## 2. Project Plan

### 2.1 Phases and Deadlines

| Phase | Main work | Deadline | Deliverable |
|---|---|---|---|
| **Planning & Management** | Proposal, plan, roles, risks, KPIs | 30 Oct 2026 | This document |
| **Lecturer Review** | Lecturer feedback, suggested improvements and approval | 2 Nov 2026 | Lecturer Review form |
| **Requirements Gathering** | Stakeholders, user stories and use cases, functional and non-functional requirements | 4 Nov 2026 | Requirements document |
| **System Analysis & Design** | Data model (ERD), logical and physical schema, architecture, DFD, diagrams, dashboard wireframes | 6 Nov 2026 | Design document |
| **Implementation** | Data cleaning and churn label, EDA, dashboard, retention report, source code on GitHub with README | 30 Nov 2026 | Code, dashboard, README |
| **Testing, Presentation & Reports** | Test cases, bug fixes, user manual, technical documentation, final slides | 4 Dec 2026 | Test report, user manual, final presentation |

### 2.2 Milestones

| # | Milestone | Date | Evidence |
|---|---|---|---|
| M1 | Proposal and plan submitted for lecturer approval | 30 Oct 2026 | This document |
| M2 | Requirements approved and design (ERD, wireframes) ready | 6 Nov 2026 | Requirements and design documents |
| M3 | Cleaned dataset with churn column ready | 13 Nov 2026 | Cleaned file and cleaning rules on GitHub |
| M4 | EDA finished and first dashboard version ready | 20 Nov 2026 | EDA notebook, first dashboard |
| M5 | Final dashboard working | 30 Nov 2026 | Final dashboard |
| M6 | All tests passed and documents finished | 4 Dec 2026 | Test report, user manual |

| # | Milestone | Date | Evidence |
|---|---|---|---|
| M7 | Final presentation | Final week of the program (Week 24) | Slides and live demo |

## 2.3 Timeline (Gantt Chart)

Shaded cells show the weeks in which each task is active. Week dates show the Monday of each week.

| Task | Owner | W1 26 Oct | W2 2 Nov | W3 9 Nov | W4 16 Nov | W5 23 Nov | W6 30 Nov |
|---|---|---|---|---|---|---|---|
| Planning & Management document | Ziad Khaled Abd El-Rahman (Team Leader) | ■ | | | | | |
| Lecturer review and approval | Ziad Khaled Abd El-Rahman (Team Leader) | ■ | ■ | | | | |
| Requirements gathering | Ahmed Ayman Naguib (Documentation Lead) | | ■ | | | | |
| System analysis & design (ERD, diagrams, wireframes) | All members | | ■ | | | | |
| Data loading, cleaning, 90-day churn label | Ziad Khaled Abd El-Rahman (Data Engineer) | | | ■ | | | |
| EDA and class-imbalance report | Omar Hassan Ibrahim (Data Analyst) | | | ■ | ■ | | |
| Churn metrics and segment tables for the dashboard | Amr Mostafa Etman (Data Scientist) | | | | ■ | | |
| Dashboard build (KPI cards, charts, filters) | Malak Ashraf Mostafa (Dashboard Developer) | | | | ■ | ■ | |
| Retention strategy report | Omar Hassan Ibrahim + Ahmed Ayman Naguib | | | | | ■ | |
| Testing, bug logging and fixing | Mennatullah Hesham Mohamed (Testing & QA) | | | | | ■ | ■ |
| User manual, final report, final slides | Ahmed Ayman Naguib (Documentation Lead) | | | | | ■ | ■ |
| Final presentation and submission | Ziad Khaled Abd El-Rahman (Team Leader) | | | | | | ■ |

### 2.4 Resource Allocation

| Resource | Used for | Responsible |
|---|---|---|
| **People** | Six team members, about 6 to 8 hours per week each, working in the roles in Section 3 | Team Leader |
| **Python, pandas, NumPy** | Loading, joining and cleaning the 9 tables; creating the churn label | Data Engineer |
| **matplotlib, seaborn, Plotly** | Charts for exploration and the final report | Data Analyst |
| **pandas aggregation (optional scikit-learn)** | Churn metrics and segment tables; optional simple model | Data Scientist |
| **Power BI Desktop or Streamlit (team chooses one)** | Building the dashboard: KPI cards, charts and filters | Dashboard Developer |
| **Jupyter Notebook / Google Colab** | Shared notebooks for exploration and analysis | All members |
| **GitHub** | Source code, data files, documents, version control | Team Leader |
| **Kaggle** | Public source of the Olist dataset (free) | Data Engineer |
| **Budget** | No budget needed: only free and open-source tools are used | Team Leader |

## 3. Task Assignment & Roles

| Role | Member | Task 1 | Task 2 | Backup owner |
|---|---|---|---|---|
| **Data Engineer** | Ziad Khaled Abd El-Rahman | Load and join the 9 tables into one customer-level data model | Clean the data, record every cleaning rule, and create the 90-day churn column | Data Scientist |
| **Data Analyst** | Omar Hassan Ibrahim | Explore churn patterns and produce the class-imbalance report and charts | Write the key customer behavior insights | Documentation Lead |
| **Data Scientist** | Amr Mostafa Etman | Calculate churn metrics and segment tables (by state, category, payment type, review score, delivery delay) | Test which factors are linked to churn and, if time allows, build the optional simple model | Data Engineer |
| **Dashboard Developer** | Malak Ashraf Mostafa | Design the dashboard layout (wireframes, KPI cards) | Build the charts and filters and test the dashboard | Data Analyst |
| **Testing & QA** | Mennatullah Hesham Mohamed | Write and run test cases (data checks, metric checks, dashboard checks) | Log bugs, check the fixes and write the test report | Dashboard Developer |
| **Documentation Lead** | Ahmed Ayman Naguib | Write and update the project documents and the retention strategy report (with input from Analyst and Data Scientist) | Prepare the user manual and the final presentation | Data Analyst |

The team has six members, within the program rule of 4 to 6. Each member owns two tasks.

### Team Leader responsibilities

The Team Leader is one of the six members (named on the first page). The Team Leader speaks on behalf of the team, delivers the project files, manages the GitHub repository, and tracks task status every week.

### Working rules

- A weekly team meeting reviews progress against the Gantt chart and updates the risk list.
- Every member commits work to GitHub regularly with clear commit messages, using feature branches and pull requests.
- Each task has a backup owner who can take over if the main owner is unavailable.
- In the final presentation, each member explains the work they completed and their own responsibility.

## 4. Risk Assessment & Mitigation

| # | Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| 1 | Severe class imbalance: most customers buy only once, so churn = 1 is the large majority | High | High | Explain the imbalance clearly in the dashboard and report, and compare churn rates between groups instead of relying on one overall number. If the optional model is built, use class weights and judge it with Precision, Recall, F1 and ROC-AUC, not accuracy alone. | Data Scientist |
| 2 | Wrong customer identifier (customer_id changes with every order) | Medium | High | Use customer_unique_id and add a data test that checks repeat customers are found. | Data Engineer |
| 3 | Recent customers cannot be fully observed for 90 days | Medium | High | Remove customers whose first order is less than 90 days before the dataset end date, and document it. | Data Engineer |
| 4 | Data leakage (using information from after the first order) | Medium | High | Use only first-order information for the breakdowns. QA reviews the logic before the dashboard is built. | Data Scientist |
| 5 | Weak or unclear patterns because the data has few customer details | Medium | Medium | Use several breakdowns (delivery delay, review score, payment type, state) and report results honestly. Even a small difference is a valid insight for the retention strategy. | Data Scientist |
| 6 | The 90-day rule is questioned or unclear | Low | Medium | Document the rule and confirm it in the lecturer review. Run the 60 and 180-day check if time allows. | Team Leader |
| 7 | Missing or messy data (null dates, reviews, product categories, duplicates) | Medium | Medium | Check data quality early and document every cleaning rule. | Data Engineer |
| 8 | Large files make tools slow (for example geolocation) | Low | Medium | Keep only the needed columns and work on aggregated customer-level tables. | Data Engineer |

| # | Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| 9 | Dashboard tool problems (setup, slow visuals) | Medium | Medium | Use pre-aggregated tables, test early on the machine used for the presentation, and keep screenshots and a recorded demo as a backup. | Dashboard Developer |
| 10 | Team member unavailable | Medium | Medium | Share task status weekly and use the backup owner for each task. | Team Leader |
| 11 | Tight deadlines | High | High | Keep the scope simple, finish core tasks before optional extras, and review progress every week. | Team Leader |
| 12 | Code or file loss | Low | High | Commit to GitHub regularly and keep all data files and documents in the repository. | Team Leader |
| 13 | Dataset is old (2016 to 2018) | High | Low | State this clearly as a limitation in the final report. | Documentation Lead |

# 5. KPIs (Key Performance Indicators)

## 5.1 Project KPIs

| KPI | Target |
|---|---|
| Tables loaded and linked correctly | 100% of the 9 tables |
| Key fields with unresolved missing values | Less than 1% |
| Churn column created with a documented rule | Done and reviewed by 13 Nov 2026 |
| Test cases passed | At least 90% |
| Deliverables submitted by their deadlines | 100% |
| Team members with documented tasks and commits on GitHub | All members |

## 5.2 Analysis KPIs

| KPI | Target |
|---|---|
| Churn label verified (repeat customers found, 90-day rule applied) | Yes, checked by Testing & QA |
| Breakdowns analyzed | At least 5 (state, category, payment type, review score, delivery status) |
| Class-imbalance report | Done and shown in the dashboard |
| Data-backed insights written | At least 5 |
| Optional simple model | If built, report Accuracy, Precision, Recall, F1-score and ROC-AUC |

## 5.3 Dashboard KPIs

| KPI | Target |
|---|---|
| Dashboard page load time | Under 5 seconds |
| Response time after changing a filter | Under 3 seconds |

| KPI | Target |
|---|---|
| Filters update all charts correctly | Yes, in all test runs |
| User adoption (ease of use) | At least 3 test users find the answers to given questions using only the user manual |

### 5.4 Business KPIs shown in the dashboard

- Overall churn rate and repeat purchase rate
- Churn rate by customer state, product category, payment type and review score
- Churn rate for on-time versus late deliveries
- Average order value and average review score
- Number and share of customers in the highest-churn segments
- Expected benefit of the proposed retention actions

## 6. Assumptions and Approval

### Assumptions

- The Olist dataset is used as downloaded from Kaggle, and the team has free access to it.
- The 90-day churn rule in Section 1.4 is accepted by the lecturer, or adjusted in the lecturer review.
- All members are available for about 6 to 8 hours per week until 4 December 2026.

### Approval

| | Name | Signature | Date |
|---|---|---|---|
| **Prepared by (Team Leader)** | Ziad Khaled Abd El-Rahman | | 30 Oct 2026 |
| **Reviewed by (Lecturer)** | Ahmed Essam Azab | | |